# PID MOTOR CONTROL LOOP

Jason Gallicchio and Claude, rev A, 2026-10-24

LAB 10L: a PID loop holds a fader (motorized potentiometer) where TARGET says.

POWER: +15 V, GND, −15 V on the POWER terminal, 1 A per rail.
    Both green lamps light. A reversed supply is blocked: the lamps stay dark.
FADER: its motor to MOTOR and GND on the MOTOR terminal.
    Its track's two ends to POT + and −, its wiper to the middle pin.
    Check with LOOP open: slide it by hand, POSITION is +7.7 V at the + end.
    LOOP closed and the slider runs to an end and stays: swap the motor leads.
INPUTS
    TARGET knob: centre detent = 0 V, clockwise +, up to +/−6.4 V.
    TARGET IN jack: a function generator, +/−12 V max.
        Try a +/−0.5 V square wave at 0.2 Hz: repeatable steps.
OUTPUTS, each through 1k: for a scope, not a load
    POSITION OUT: the fader's wiper, buffered.  DRIVE OUT: the motor drive.
SWITCHES
    TARGET SOURCE  1 = the knob
                   3 = the TARGET IN jack
    LOOP           1 = open: no feedback; DRIVE = P GAIN x TARGET (+ D, I if on)
                       P GAIN anticlockwise, D and I off: DRIVE = TARGET
                   3 = closed: POSITION fed back, the loop holds TARGET
    D IN / OFF     1 = off: the differentiator's output shorted
                   3 = D into the sum
    D CAPACITOR    1 = 22 nF, for this fader
                   3 = 100 nF, for the book's slower rotary pot
    I IN / OFF     1 = off: the 10 uF shorted, so I restarts from zero
                   3 = I into the sum
TRIMMERS: clockwise = more resistance: more P, more D, but LESS I.
START: every switch on 1.  P and D GAIN anticlockwise, I GAIN clockwise.
    LOOP to 3: the slider follows the knob.  Raise P GAIN until it rings.
    D IN, raise D GAIN: the ringing dies.  Lower P; I IN: the error goes to 0.
WARNINGS
    A stall opens the 0.35 A fuse in seconds; red lamps; it resets itself.
    Never short MOTOR to GND or to a rail.  Power off to wire the fader.

https://github.com/gallicchio/motor-pid
circuit, scripts, schematic, this board

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